Welcome to Your

Full Stack Software Development Program

Build Code. Create Projects. Start Your Tech Career.

COURSE WELCOME E-KIT

This editable e-kit gives learners the roadmap, outcomes, checklist, class guidelines, starter concepts, and support details before the first session.

Mode	Classroom + Online support available
Best For	Students, beginners, graduates, career switchers, and working professionals
Outcome	Build foundations in programming, web development, databases, Java frameworks, and AI/ML basics

A Warm Welcome

Welcome to Expert Computer Academy. You have taken a smart step toward building a practical, job-ready software development skill set. This program is designed to help you move from basic computer knowledge to structured programming, web development, database handling, Java frameworks, and emerging AI/ML concepts.

By the end of this course, you should be able to think logically, write code confidently, understand databases, build web pages, work with backend concepts, and explain your projects clearly during interviews or career discussions.

What You Will Gain

- Confidence in programming fundamentals and logic building
- Hands-on exposure to C++, Python, Java, HTML5, MySQL, JSP, Servlet, Hibernate, Spring, and AI/ML basics
- Ability to build structured mini projects and explain your work professionally
- Better preparation for internships, fresher developer roles, and advanced technical learning
- A stronger foundation for full stack development, backend development, and software engineering roles

Software development is not about memorizing code - it is about learning how to think, solve problems, and build useful solutions.

Course Curriculum Roadmap

Module 1	**Logic Building using Flowcharts & Pseudocode** Learn step-by-step thinking, flowcharts, pseudocode, dry runs, and algorithm planning.
Module 2	**Object-Oriented Programming using C++** Understand classes, objects, constructors, inheritance, polymorphism, and practical OOP examples.
Module 3	**Web Designing with HTML5** Create structured pages using HTML5, semantic tags, forms, tables, media, and clean layouts.
Module 4	**Python Programming** Learn Python syntax, variables, loops, functions, files, collections, and basic automation.
Module 5	**Relational Database Management System - MySQL** Design databases, write SQL queries, use joins, filters, grouping, and manage relational data.
Module 6	**Data Structures and Algorithms** Build DSA thinking using arrays, strings, stacks, queues, linked lists, searching, and sorting.
Module 7	**Programming with Java** Strengthen OOP through Java syntax, classes, exceptions, collections, and application coding.
Module 8	**Web Application Development with JSP, Servlet, JSF, Hibernate and Spring Framework** Understand Java web apps, request-response flow, MVC, persistence, and framework-based development.
Module 9	**Artificial Intelligence & Machine Learning** Explore AI/ML basics, data thinking, model concepts, and modern software use cases.

Practical Focus: Every topic should be supported with coding practice, assignments, examples, and mini-project work so learners can apply concepts immediately.

Learning Outcomes

Think like a programmer Break problems into steps using flowcharts, pseudocode, and logical reasoning.	**Write clean code** Practice syntax, functions, OOP concepts, and structured programming habits.
Build web foundations Create clean HTML5 pages and understand how web applications are structured.	**Work with databases** Use MySQL to store, retrieve, filter, and manage structured data.
Understand Java web concepts Learn the basics of JSP, Servlet, Hibernate, Spring, and application flow.	**Prepare for interviews** Explain projects, logic, DSA basics, database queries, and programming concepts clearly.

Mini Portfolio Projects You Can Build

- Student Management System
- Personal Portfolio Website
- Login and Registration Flow
- Library or Inventory Management System
- Basic Java Web Application
- AI/ML Concept Demo or Use-Case Presentation

Before Your First Class

Use this checklist to come prepared and get maximum value from every session.

☐ Bring a notebook or digital notes app for logic, syntax, commands, and project steps.

☐ Carry your laptop if available and keep it charged for practical coding sessions.

☐ Install basic software as instructed by the academy, such as code editor, compiler, browser, and database tools.

☐ Create a folder named Full Stack Practice to save class files, assignments, and projects.

☐ Practice typing simple programs before class, such as print statements, variables, loops, and conditions.

☐ Ask questions during practice - programming improves fastest when doubts are solved immediately.

Recommended Learning Discipline

Daily practice	30-45 minutes after every class
Weekly revision	Revise syntax, logic, database queries, and previous assignments once a week
Project habit	Save every class task as a portfolio sample with proper file names
Doubt clearing	Ask doubts before moving to the next module
Consistency	Avoid long breaks because coding confidence grows through regular practice

Class Guidelines & Support

During Class	After Class
• Join on time and keep your code editor ready • Follow the instructor step by step before trying shortcuts • Do not skip hands-on practice tasks • Ask doubts when logic, syntax, or errors are unclear	• Complete the given coding assignment • Save your work with proper file names • Revise the logic, not just the final output • Share errors with screenshot or file reference

Support Details

Phone	7282983335 | 9128812325
Address	Kumar Tower, 2nd Floor, Boring Rd Crossing, Patna-800001, Bihar
Online Course	Online courses also available
Academy	Expert Computer Academy - 38 Years of Excellence
Editable Fields	Batch timing, trainer name, course duration, fee, and registration link can be updated before sharing

Starter Coding Cheat Sheet

Keep this page handy during the first few sessions. The instructor will explain each concept with practical examples and coding tasks.

Concept	Purpose	Example
Variable	Stores a value in memory	int age = 20;
Condition	Runs code based on a decision	if (marks >= 50) { pass; }
Loop	Repeats a block of code	for (int i=1; i<=5; i++)
Function	Reusable block of code	int add(int a, int b)
Array	Stores multiple values	int marks[5];
Class	Blueprint for objects	class Student { ... }
SQL SELECT	Reads data from a table	SELECT * FROM students;
HTML Form	Collects user input	<form>...</form>

Tip: Do not memorize code blindly. Understand what problem each concept solves, then practice it on different examples.

Your 30-Day Full Stack Growth Plan

Timeline	Focus Area
Week 1	Logic building + flowcharts + pseudocode + basic programming practice
Week 2	C++ OOP + HTML5 structure + Python basics and assignments
Week 3	MySQL queries + DSA basics + Java programming practice
Week 4	Java web concepts + mini project + AI/ML awareness + interview revision

Final Message

Software development becomes powerful when you practice consistently. Attend each session, complete your assignments, and build your own mini project. Your goal is not only to finish the course - your goal is to become confident enough to write, debug, and explain code in real work.

Ready to begin? Save this e-kit, attend your first class, and start building your software development confidence from Day 1.